Mohammad Sheriff Mehmood

Senior Data Scientist · NLP & Generative AI Engineer

+91 7999937693 · mdsheriff2702@gmail.com · [LinkedIn]() · [github.com/sheriff786]() · [Portfolio]()

PROFESSIONAL SUMMARY

Senior Data Scientist & AI/ML Engineer with 6+ years of experience building scalable AI-powered applications. Specializing in NLP · LLMs · RAG systems · and production MLOps, I have shipped enterprise-grade GenAI systems that delivered $1M+ revenue uplift · cut churn by 40% · and accelerated model iteration by 60% across healthcare · finance · and retail. Experienced in cross-functional collaboration with product · engineering · and operations teams. Open-source author: published authentica on PyPI, actively used in production at Businessolver (Healthcare Benefits SaaS).

TECHNICAL SKILLS

Languages: Python (primary) — proficient in TensorFlow · PyTorch · Scikit-learn · NumPy · Pandas · OpenCV · YOLO · CNN

GenAI & LLMs: LangChain · LangSmith · OpenAI API · Gemini · RAG · LLMOps · Fine-tuning · Prompt Engineering · Agentic AI · MCP · Transformer Models · Sentence-Transformers · CrewAI

MLOps & Cloud: MLflow · Docker · Kubernetes · Airflow · GCP (Vertex AI) · AWS (SageMaker · ECS · CodePipeline) · CI/CD

Databases: PostgreSQL · MongoDB · SQL · ChromaDB · FAISS · Pinecone

Tools & Other: Tableau · Weights & Biases · FastAPI · Flask · Django · A/B Testing · REST API Design · Git

PROFESSIONAL EXPERIENCE

ValueLabs · Hyderabad, India

Senior Data Scientist | AI/ML Engineer

Mar 2025 – Present

Embedded with Businessolver's Data Science team as a full-stack ML Engineer; designed and shipped an automated document analysis pipeline leveraging computer vision · VLMs · and AWS-native tooling to process healthcare documents at scale.

- Document Analysis Pipeline: Designed an end-to-end multi-modal pipeline on AWS (Comprehend · Textract · S3 · ECS) to classify and extract data from 47,000+ healthcare documents · significantly cutting manual claim validation time.
- Query-Based Document Understanding: Built a custom QA system integrating visual features from Textract OCR with textual embeddings across varied layouts; deployed inference on AWS Lambda achieving 93% extraction precision.
- Annotation & Monitoring UI: Built an interactive React UI with REST API for data annotation/validation · real-time model metric visualisation · performance monitoring · and data-driven threshold tuning streamlining stakeholder review cycles.
- Stacking Ensemble Model: Trained and tuned a stacking ensemble (Random Forest · XGBoost · LightGBM) via cross-validation achieving 0.78 F1 and 0.81 AUC; applied SHAP and Gini impurity for model explainability.

LTIMindtree · Pune, India

Senior Data Scientist – NLP Engineer

Jan 2024 – Mar 2025

- GenAI Chatbot & RAG System: Architected production-grade RAG chatbot (Gemini + FAISS Vector DB) serving 15+ team members with 90% query accuracy and sub-2s response time for enterprise knowledge retrieval.
- LLMOps Pipeline: Built end-to-end LLMOps pipeline using OpenAI · LangChain · ChromaDB · and sentence-transformers cutting model iteration time by 60% and reducing deployment overhead by 50%.
- Healthcare Revenue Model: Deployed anomaly-detection targeting model generating $1M+ in annual healthcare revenue at 90% precision; reduced false positives by 20% via class-weighted ensemble (Logistic Regression + Random Forest).
- Workflow Orchestration: Designed Airflow DAGs to automate multi-step ML pipelines · cutting end-to-end processing time by 40% and improving delivery velocity across a 7-member team.
- Semantic NLP (BioBERT): Generated vector embeddings for ICD-9 and medical procedure codes using BioBERT · improving semantic search accuracy by 18% over TF-IDF baseline.

Cognizant Technology Solutions · Chennai, India
Data Scientist
Dec 2019 – Jan 2024

Joined as Programmer Analyst → core ML systems, promoted to Data Scientist (leading cross-functional AI/NLP initiatives).

- ML Model Development: Built and deployed supervised ML models improving prediction accuracy by 20% and saving $200K in operational costs; improved baselines by 5–10% using Neural Networks · ensemble methods · GridSearchCV.
- AI Customer Support Platform: Led 7-member cross-functional team to build NLP-powered support application; reduced customer churn by 40% and generated $300K additional quarterly revenue through context-aware intelligent responses.
- Fraud Detection System: Reduced false-positive fraud alerts by 20% using Logistic Regression and Random Forest with class-weighted resampling — received company Innovation Award.
- Data Platform & BI: Unified 10+ heterogeneous data sources (databases · APIs · third-party feeds) into ML-ready pipelines; built Tableau dashboards tracking 20+ KPIs · improving cross-team reporting efficiency by 30%.

RESEARCH PROJECTS

Abstractive Text Summarisation | Python · BERT · NLP · Transformers View on GitHub →

- Applied BERT and NLP toolkits to summarise live chat context, improving agent efficiency by 35% and reducing customer response times by 25%.
- Gave agents instant access to previous chat history — reducing handle time and improving resolution quality.

End-to-End Topic Modelling & MLOps on AWS | Python · LDA · Docker · AWS ECS · MLflow · CodePipeline · EC2 View on GitHub →

- Built LDA topic modelling pipeline on AWS ECS with blue/green deployment — cut processing time by 40% and deployment time by 60%.
- Automated full CI/CD (CodeCommit · CodeBuild · CodeDeploy · CodePipeline); reduced manual effort by 70%. Used MLflow for model versioning.

Visa Approval Forecasting (US Immigration) | Python · ML · MongoDB · Docker · AWS · GridSearchCV View on GitHub →

- Predicted US visa approval outcomes using production-grade ML models; performed EDA · feature engineering · and hyperparameter tuning via GridSearchCV.
- Connected MongoDB for data persistence; containerised solution with Docker for AWS deployment.

Disease Classification Mobile App | TensorFlow · CNN · TFLite · FastAPI · GCP · ReactJS · React Native View on GitHub →

- Achieved 92% accuracy; quantized CNN with TF Lite — reduced model size by 80% and inference latency by 50%.
- Deployed serverless inference on GCP Cloud Functions (25% cost reduction); served predictions via FastAPI to React frontends.

Accident & Fall Detection System | Python · YOLO · OpenCV · CNN · CUDA · SMTP View on GitHub →

- Built real-time human fall detection system achieving 95% accuracy using YOLO + CUDA toolkit on edge hardware.
- Designed smart camera surveillance for highway accident detection; automated ambulance dispatch via SMTP alerts · cutting emergency response time by 40%.

EDUCATION

M.Tech in Data Science & Engineering · BITS Pilani, India · Oct 2021 – Oct 2023
B.E in Computer Science & Engineering · OIST Bhopal, India · Aug 2015 – Jun 2019

CERTIFICATIONS & AWARDS

- GCP Professional ML Engineer (Certified Feb 2023)
- Certified MLOps Developer – Dataiku
- IBM Professional Data Science Certification
- 1st Prize – Trizetto Hackathon (Patient Data Integration · 95% approval rate)
- OpenAI Whisper Hackathon – Hearing-impaired accessibility app
- Secretary · Toastmasters Club · Speaker · PyData Conference